

AGRICULTURE: INDIA'S JOURNEY FROM SURVIVAL TO SUPREMACY - AND THE ROAD AHEAD

India's economic rise is one of the defining stories of the 21st century, and at the heart of this transformation lies a force that has sustained our civilization for millennia—agriculture. From the fertile plains of the Indus and the Ganga to the modern, technology-enabled farms of today, Indian agriculture has travelled a long and inspiring journey. It is this sector that feeds 1.4 billion Indians, provides livelihoods to more than half of our population, and anchors rural India, which remains the backbone of the nation's social and economic stability.



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The modern transformation of Indian agriculture began with the **Green Revolution of the 1960s**. At a time when India depended on **PL-480 food imports from the USA**, visionary leadership and scientific excellence changed the course of history. High-yielding seeds, improved irrigation and fertiliser use helped achieve food self-sufficiency. At the same time, the **White Revolution** made India the world's largest milk producer and empowered millions of small farmers through cooperatives. These were not just agricultural milestones but **nation-building achievements** that built confidence, dignity and long-term growth in crop production. Today, India stands as a global agricultural powerhouse. We have become the world's largest producer of rice, overtaking China, with production exceeding 150 million tonnes. Our rice exports have surged, helping feed populations across

Asia, Africa and the Middle East. India is also among the leading producers of fruits, vegetables, milk, spices, cotton and sugar. Behind each of these achievements stand our farmers, scientists and industry—their collective contribution to national development is truly remarkable. Our farmers work under some of the most challenging climatic and market conditions in the world, while our scientists continuously innovate new seeds and agronomic practices suited to India's diverse agro-climatic zones. Yet, despite these impressive achievements, a deep paradox remains. India's agricultural potential far exceeds its current performance. While we are global leaders in volume, our per-hectare productivity and farmer incomes remain well below those of many developed and even some developing countries. China's agricultural GDP is nearly three times that of India, despite both countries having comparable land

and labour resources. This gap exists not because Indian farmers lack capability or commitment, but because systemic constraints prevent our agriculture sector from unlocking its full potential.

One of the most serious constraints is limited access to modern technology, particularly for small and marginal farmers, who make up nearly **85% of the farming community**. With highly fragmented landholdings, most farmers struggle to access advanced machinery, precision tools, high-quality seeds and modern crop protection solutions. During the recent **Viksit Krishi Sankalp Abhiyan**, Union Agriculture Minister **Shri Shivraj Singh Chouhan** noted that many of the latest technologies have yet to reach this segment. Without technology, productivity stagnates, costs rise, and farmers remain vulnerable to climate and market risks. Even more damaging is the widespread prevalence of

